

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY 2026 6 QUESTIONS 10% WEIGHTAGE ↓24.98%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 66 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Current Electricity - Core Application	66 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Current Electricity - Core Application (Very High Yield — 66 PYQs)

Core Principle: Key problem pattern in Kcet Current Electricity.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure, the values of currents I_1 , I_2 and I_3 respectively are..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following circuits is correct for verification of Ohm's law?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure, the values of currents I_1 , I_2 and I_3 respectively are

- (A) 6 A, 1.5 A and 1 A
- (B) 4 A, 2.5 A and 2 A
- (C) 4 A, 2.5 A and 1 A
- (D) 6 A, 4.5 A and 1.5 A

Correct Answer: None

Solution Analysis: At any junction, Kirchhoff's Current Law says that the total current entering the junction is equal to the total current leaving it. For the first junction, 5A enters. The currents leaving are 1A and I_1 . So, $5 = 1 + I_1 \Rightarrow I_1 = 5 - 1 = 4A$ For the second junction, I_1 enters. The currents leaving are 1.5A and I_2 . So, $I_1 = 1.5 + I_2$ Putting $I_1 = 4A$, $4 = 1.5 + I_2 \Rightarrow I_2 = 4 - 1.5 = 2.5A$ For the third junction, I_2 enters. The currents leaving are 1.5A and I_3 . So, $I_2 = 1.5 + I_3$ Putting $I_2 = 2.5A$, $2.5 = 1.5 + I_3 \Rightarrow I_3 = 2.5 - 1.5 = 1A$